

Audiometry

"Audiometry: measurement of the range and sensitivity of a person's sense of hearing". -Oxford Languages Dictionary

Scientific discourse has sought ways of quantifying the subjective sense of hearing. Audiometry is one means of approaching that goal. Emerging alongside telephone technology, the audiometer allows an examiner to present a series of tones to an examinee in order to record if they are heard. Importantly, audiometry tests hearing outside of any real-world context—it should be understood as just one way to think about hearing.

In small groups, take turns playing the roles of examiner and examinee using a vintage Belltone 109 Audiometer. If you do not want to be an examinee, this role is optional.

Keep in mind that the results show how well you have heard this particular piece of equipment in a very uncontrolled environment that has little to do with hearing in the real world. The point is that hearing is variable.

Set the audiometer controls as follows

1. Turn on the audiometer and allow it to warm up for at least 3 minutes
2. Verify that the headphone jacks are correctly inserted into their respective ports
3. Set Tone mode to PULSED (if unavailable, choose NORM OFF)
4. Randomly pick output as either RIGHT or LEFT ear (do not inform examinee of your choice)
5. Set the frequency to 1000 Hz and the dB dial to 30 dB

Procedure for Hearing Level Test

1. Have the examinee face away from the machine/the examiner, and give them the response button (if available—otherwise, the examinee should raise their finger).
2. Present the tone by pressing the TONE pad for about 1 second (about 3 pulses). If the examinee does not respond, increase the intensity in 15 dB steps until a response is obtained.
DO NOT EXCEED 65 dB.
3. When a positive response is obtained from the examinee, drop the level in 10 dB steps and present the signal again until no response is obtained.
4. When there is no response, increase the intensity in 5 dB steps and present the signal each time until the tone can again be heard by the examinee. Count this dB response as the threshold.
5. Record threshold on the Audiometry Chart for the respective ear.
6. Continue to search for the threshold in this manner for the following frequencies (125, 250, 500, 1000, 2000, 4000, 8000) decreasing the stimulus in 10 dB steps following each positive response and increasing the stimulus in 5 dB steps following each nonresponse. Always begin with an initial intensity of 30 dB HL or intensity level 15 dB above the threshold at the preceding frequency, whichever is greater.
7. When thresholds have been obtained at all frequencies in the first test ear, press the RIGHT or LEFT button to shift to the other ear. Conduct the test in this ear in exactly the same way.

A procedure based on https://www.cdc.gov/nchs/data/nhanes/nhanes_03_04/au.pdf. Modified for operation on the Beltone MODEL 109 Audiometer by Zac Watson.